

Comparing Current Take-Off Data of Cessna 177 RG to Expected Results in 1976 Pilot's Operating Handbook

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May-June 2026

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1 Introduction

In this report, we will be looking at Mike Montgomery’s personal flight logs and comparing them to what his Pilot’s Operating Handbook (POH) predicts these measurements will be. The POH is a handbook provided by the aircraft manufacturer containing standardized performance predictions for a specific aircraft model under various conditions. For this study our aircraft is a *Cessna 177 RG*. While flying has a lot of variables, we are specifically looking at the distance in feet his aircraft rolled on the runway before it took off, and the total distance it took him from the start of rolling to being able to fly over a 50’ obstacle. For those interested, citations to the illustrations and POH handbook are provided at the end of this report.

This analysis uses 16 flight records collected by the pilot between January 2025 and May 2026. For each flight, the pilot recorded environmental conditions including temperature, pressure altitude, and gross weight, and manually calculated the expected ground roll and 50ft clearance distances using the POH performance tables. Actual ground roll and 50ft clearance distances were also recorded for each flight using a Garmin GNC 355 GPS (a navigation system widely used in general aviation), allowing for a direct comparison between predicted and actual performance. The table Montgomery filled out as well as the POH tables with definitions can be found starting in Section 8 of this report.

This first illustration shows the total distance from starting to roll to then clearing a 50’ obstacle.¹ The second shows the differences in “climbing” at different angles.² As we can see, the angle at which a pilot climbs after lift-off makes a significant difference in how quickly they can clear a 50’ obstacle.

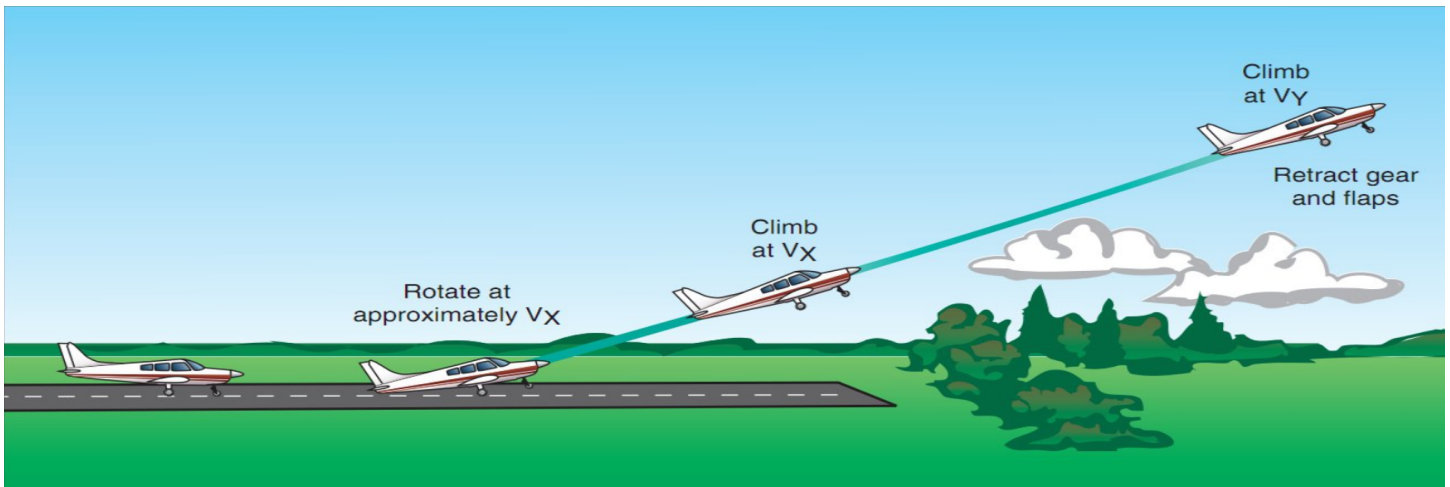


Figure 1: take-off sequence showing ground roll and climb out

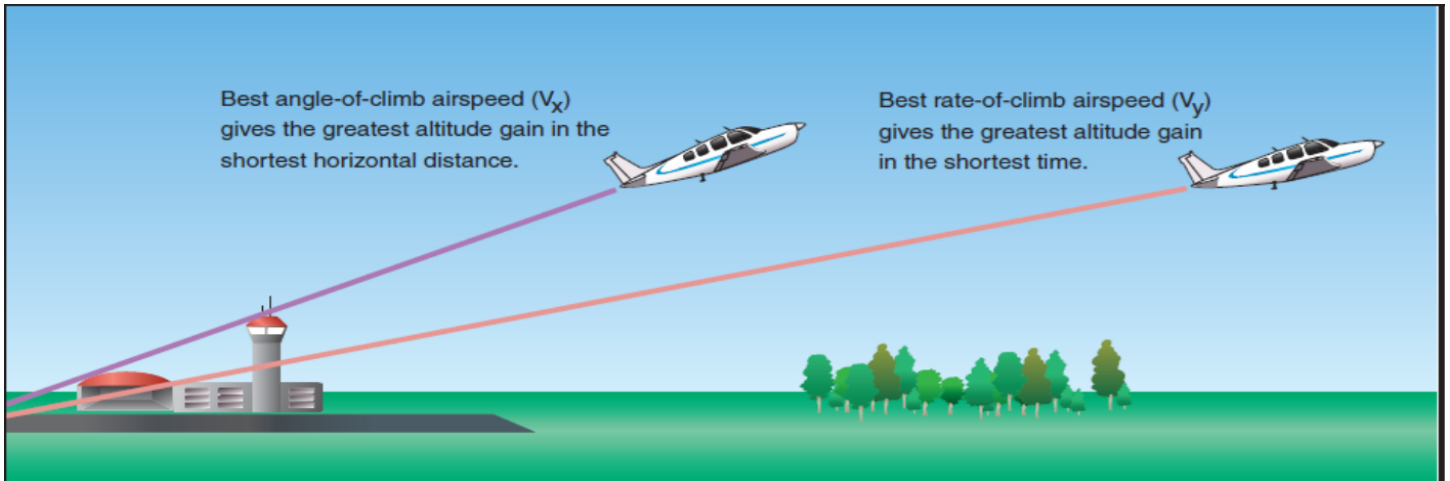


Figure 2: Climb angle (V_x) vs climb rate (V_y) tradeoff

With these concepts in mind, we can now move into the statistical analysis. To assess whether the pilot’s performance differed significantly from POH predictions, we ran paired t-tests comparing computed and actual distances for both ground roll and 50ft clearance. We then used a series of visualizations: scatterplots, histograms, Bland-Altman plots, and time series plots, to further explore the nature and consistency of any differences found.

This analysis was conducted using *R*, a free, well known programming language for statistical reports.

2 Paired t-test

In statistics, *paired t-tests* are used to compare two measurements from the same subject to see if the average difference between them is significant. The main output we get from a t-test is a *p-value*.

P-values tell us how likely we would be to observe the difference we see in our values if there was truly no difference between what we're comparing. So a high p-value tells us we're very likely to see these numbers if we assume what we're comparing is equal to each other, and a low value would tell us we're *not* likely to see this difference if we assume the two groups are equal to each other.

Typically in statistics if a p-value is less than 0.05, or 5%, we determine the differences in our averages are significant and our two groups are *not* aligned with each other.³ If our p-value is equal to or above 0.05, we say we don't have enough evidence to prove the two groups are different. In this case, we're asking "Are this pilot's performances different than what is predicted for him?" and our p-value will help us determine either yes (p-value of below 0.05) or no (p-value of 0.05 or greater). It's also worth noting we're not technically proving if he "is" flying within the measurements, we're only saying we can't prove he isn't. In statistics we'd say our *null hypothesis* is, "our pilot's performances are not different from what the POH predicts," and we will either *reject* or *fail to reject* that hypothesis.

Knowing that, let's run two separate t-tests: one comparing the rolling distances and another comparing the distances until the plane can clear a 50' obstacle. The tables we'll be referencing are found in Section 8 of this report.

Please note: The following results for this test have output from my coding. The definitions for *flightData*, *TO Roll*, *TO Dist*, *50' Dist* and *50' Obs* can be found in Section 8 for those curious. For those strictly interested in the results, however, we'll only look at the *p-value* and *mean difference* values.

2.1 T-test for Rolling Distance

Paired t-test

```
data: flightData$`TO Roll` and flightData$`TO Dist`
t = -1.2394, df = 15, p-value = 0.2342
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 -102.84012  27.21512
sample estimates:
mean difference
 -37.8125
```

Here we look at our p-value of 0.2342, or 23.42%. This is greater than our 5% standard, so we can conclude we don't have enough evidence to say the pilot is performing differently than predicted. Simply put, he's performing as predicted. We can also see the mean difference at the bottom of the report is -37.81, this means on average the pilot's actual ground roll is 37.81 feet shorter than predicted. Not too bad! Now let's look at the distance to clear 50' obstacles.

2.2 T-test for Distance to Clear 50'

Paired t-test

```
data: flightData$`50' Dist` and flightData$`50' Obs`
t = 4.1383, df = 15, p-value = 0.0008757
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 166.7594 520.9906
sample estimates:
mean difference
 343.875
```

Here our p-value is 0.0008757, or 0.09%. This is extremely low, much lower than our 5% standard. Simply, this is saying the odds of us seeing this difference in our means if our computed and actual results were aligned is 0.09%, basically zero! So something is misaligned here. This means we have more than enough evidence to conclude the pilot is not performing in accordance to his predicted outcomes. Looking at our mean difference at the bottom of the output, we see 343.875. This means the pilot is on average reaching the altitude to clear a 50' obstacle about 343.88 feet *after* he is predicted to.

3 Plots for Distances

Here we will be seeing how take-off data differed from what was predicted. The closer a point is to the red line, the closer it was to being the same as what was predicted. This will help us see if there are any trends in how we deviate from what is predicted, if at all. We'll also have a blue line shaped from our observations to clearly see any trend they have, which we can compare to the red line.

3.1 Ground Roll

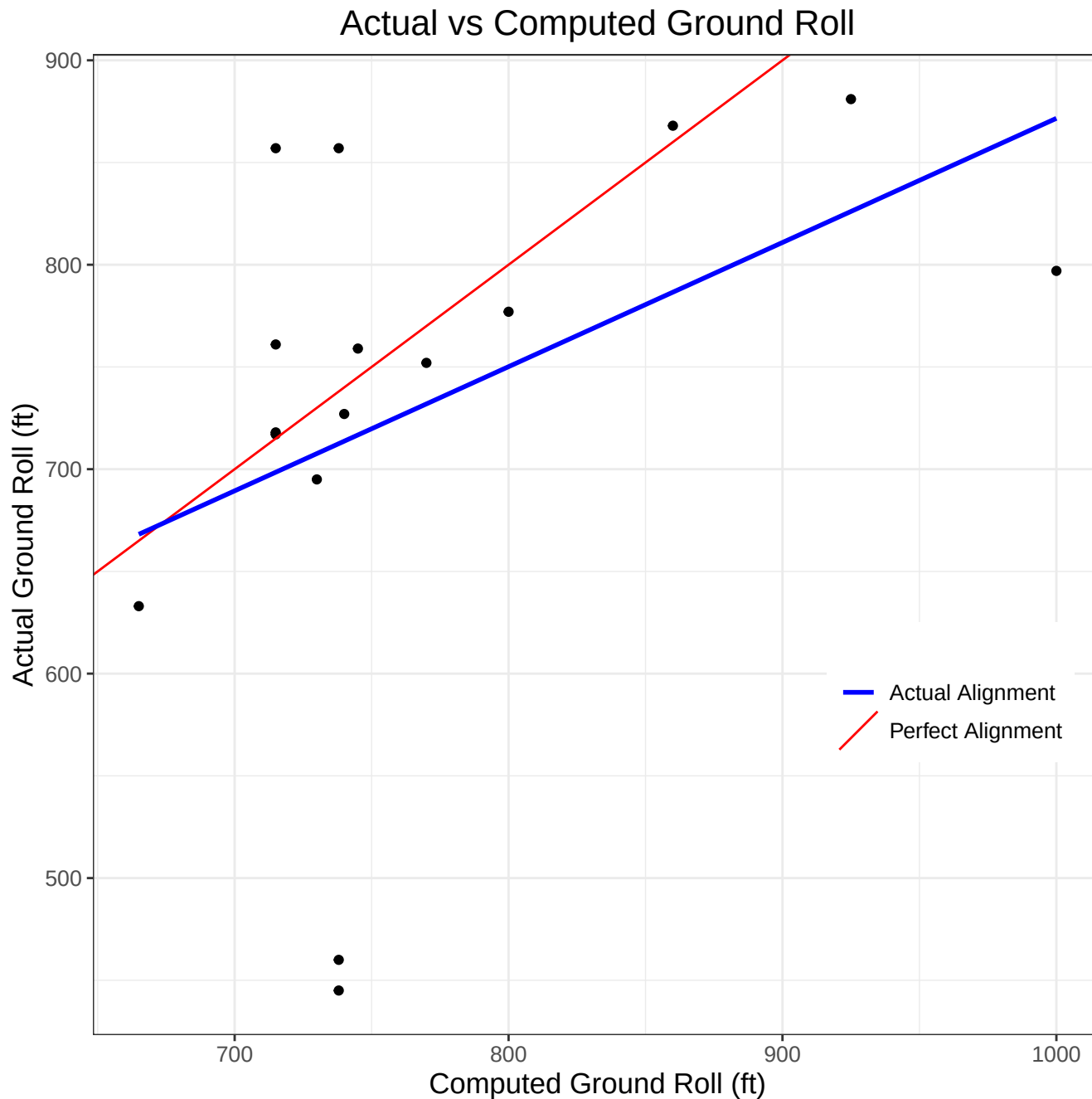


Figure 3: Actual vs Computed Ground Roll Distance

This first graph is pretty typical, we see points on either side of the line, some closer than others, with a few outliers here and there. We can also see two flights at the bottom of the graph (about 750' computed, about 450' actual) where it seems the pilot took off significantly earlier than predicted.

Our blue line is the average line our actual points roughly follow, and we can see it starts similar to our predicted, and as computed distances increase, the pilot's actual ground roll tends to fall slightly below the predicted.

3.2 50' Distance

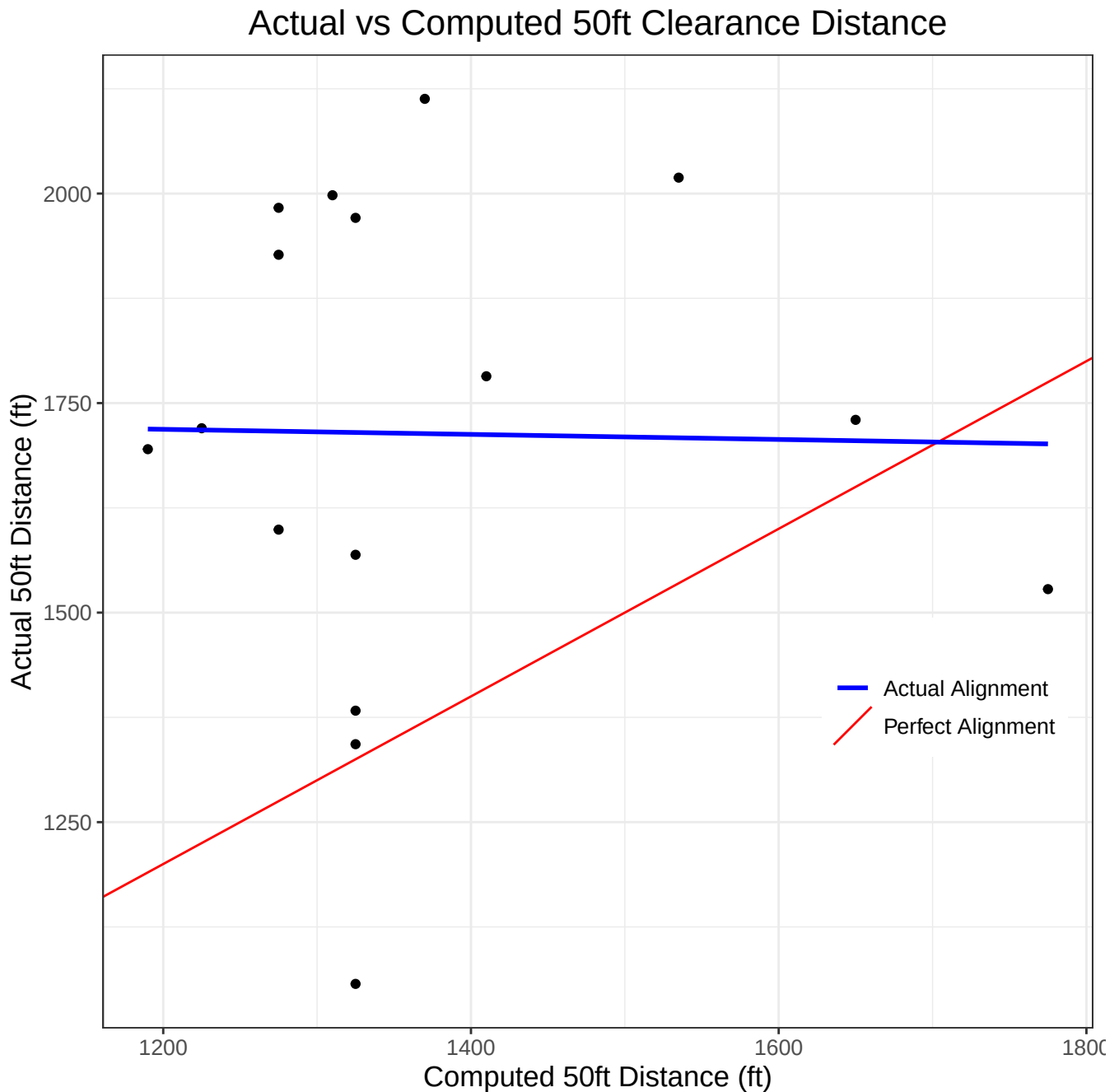


Figure 4: Actual vs Computed 50ft Clearance Distance

For this graph we can see right away almost all our points are above the red line, and not too many are near it. Our blue line shows the pilot averaged a little less than 1750' on each take-off. This is where statistics can be misleading. We see the pilot doesn't actually have that many clearance distances at the 1750' mark, however he does have a lot above, near the 2000' range, and below scattered down. However, even though our "average" can be misleading, the main takeaway from this is our blue line is drastically different from our red one. In addition to this, the blue line doesn't have that much of a slope, meaning the pilot's performance stays almost the same regardless of what the POH predicts. The right-most point on the graph is an outlier, clearly examined in section 7.2.

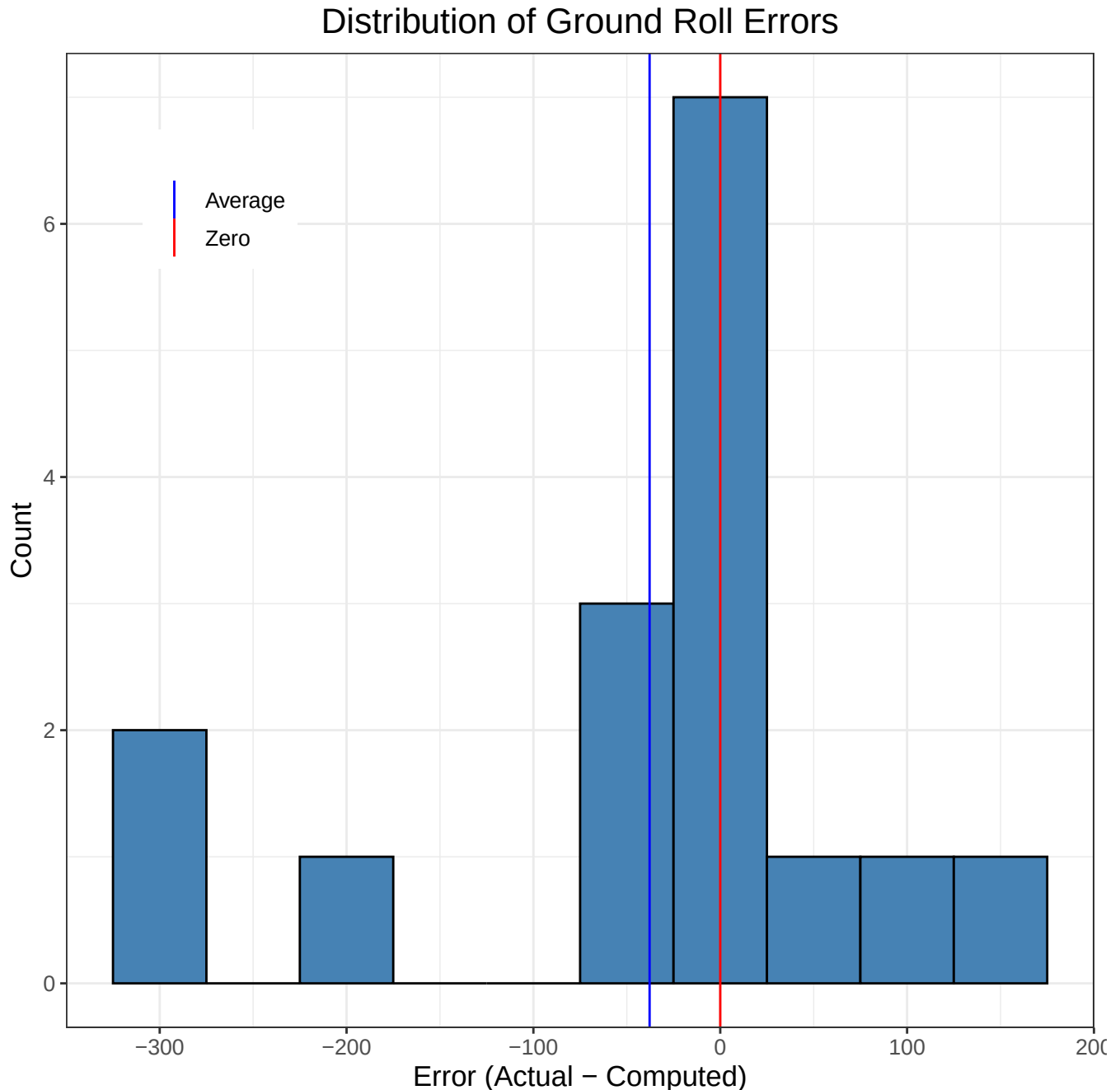
Looking at both plots together, we see something interesting. Although the pilot stayed pretty aligned with the POH during the Ground Roll, the distance to make it to 50' was almost consistently off by hundreds of feet. This could mean the 50' performance may be driven by something else the POH may not account for, such as engine/aircraft performance or angle of climb, as opposed to the environmental conditions like weight and temperature.

4 Observing Deltas

Here we're going to organize the *deltas* (differences between actual and predicted distances) to get a better visual of how the pilot is operating. We will group the deltas with certain distances, then count the amount in each group. We'll also see a red line to better mark our predicted value, and a blue line to mark the average of our observations.

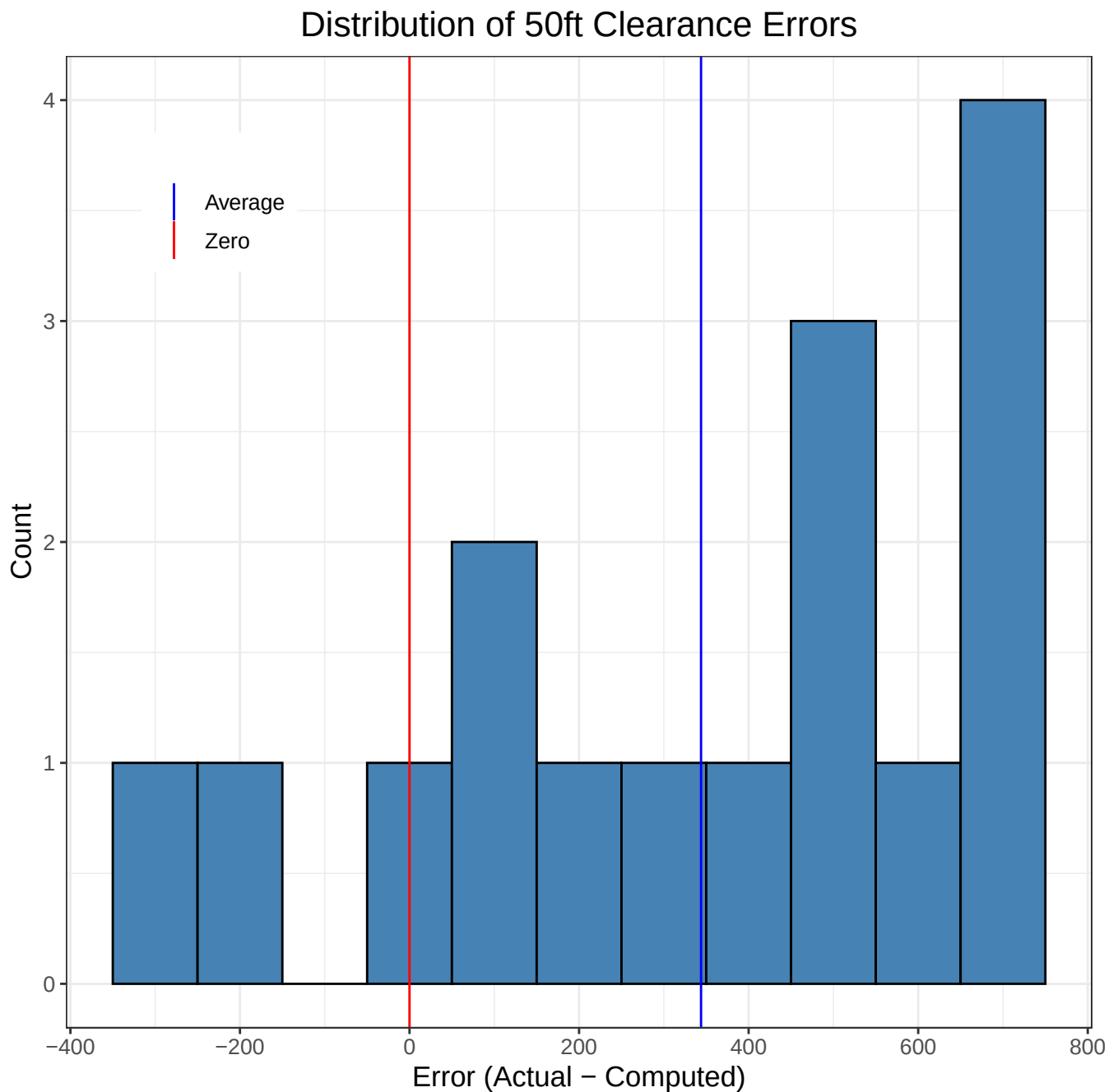
For the Ground Roll deltas we'll have groups of 50 feet and for our 50' Obstacle deltas we'll have groups of 100 feet. The closer our deltas are to zero (0), the closer it is to our predicted value.

4.1 Ground Roll



Here we can see the pilot has the majority of his flights right in line (pun intended) with what's predicted for him. We see some of the outliers we pointed out before on the left side, where the pilot lifted off earlier than predicted. Most flights cluster near zero with a few outliers pulling the average to the left.

4.2 50' Obstacle

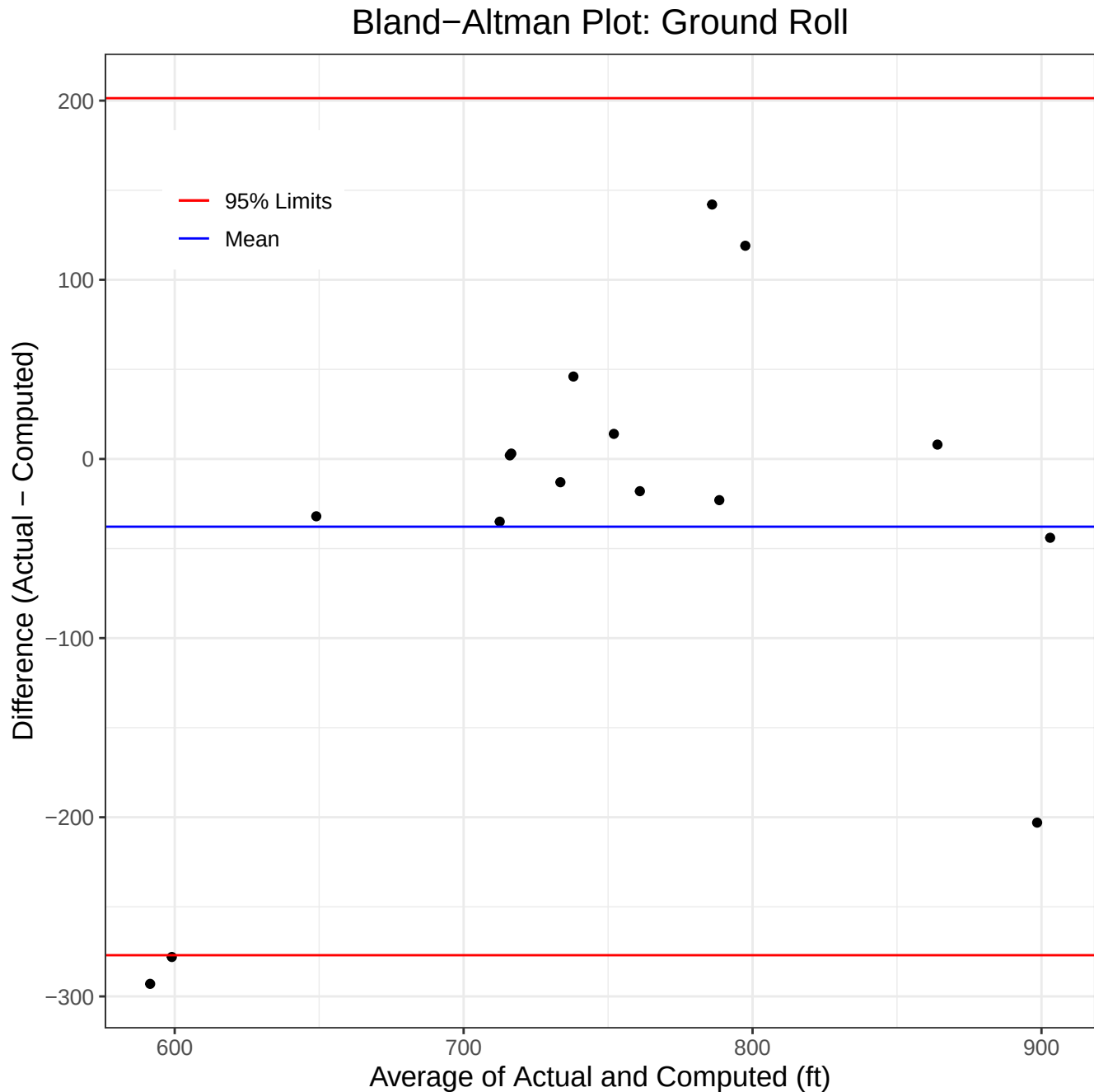


Here we can see how frequently the pilot took extra distance to clear a 50' obstacle. We see most of our differences are from 600 to 800 feet! That's pretty substantial. Although we have some pile up towards the end, we see there's no gaps on the right side of the graph, meaning the amount of overrun varies quite a bit flight to flight. We also see our average is much higher than our predicted, which confirms what we saw earlier.

5 Bland-Altman

Next we're going to use a *Bland-Altman* graph. This graph will have the *average* of the predicted and actual of each flight on the x-axis, and the *difference* between the two on the y-axis. This is to show us if there's any correlation between the differences shown. The blue line we'll see is the average between the flights, and the red lines are where we'd predict 95% of differences (or *two standard deviations*, for appropriate statistician lingo) to fall. This is important when determining if a flight is especially noteworthy.

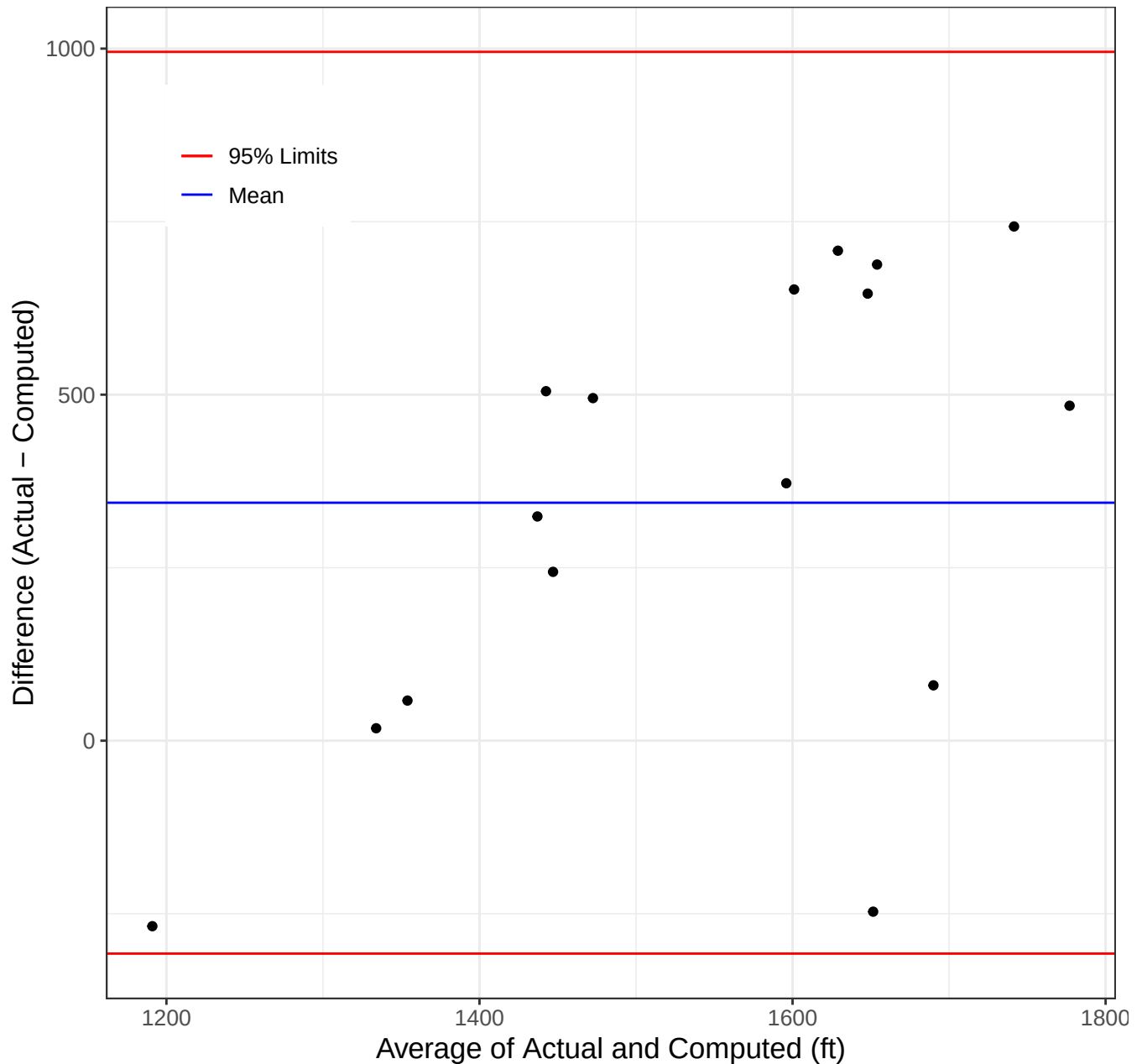
5.1 Ground Roll



For the ground roll, we see most of the flights are clustered near the blue line with the majority falling into a reasonable range. We also see on the bottom left we have *two* flights outside our red lines. This means these two flights are outliers, and are also the main reason our average is being brought down. Our range of the flights is about -300' to +200', around 500' of variation between all the flights. There's also no obvious trend or pattern between the points, so we can see the difference between the computed and actual stays almost equal throughout different take-off distances.

5.2 50' Obstacle

Bland–Altman Plot: 50ft Clearance Distance



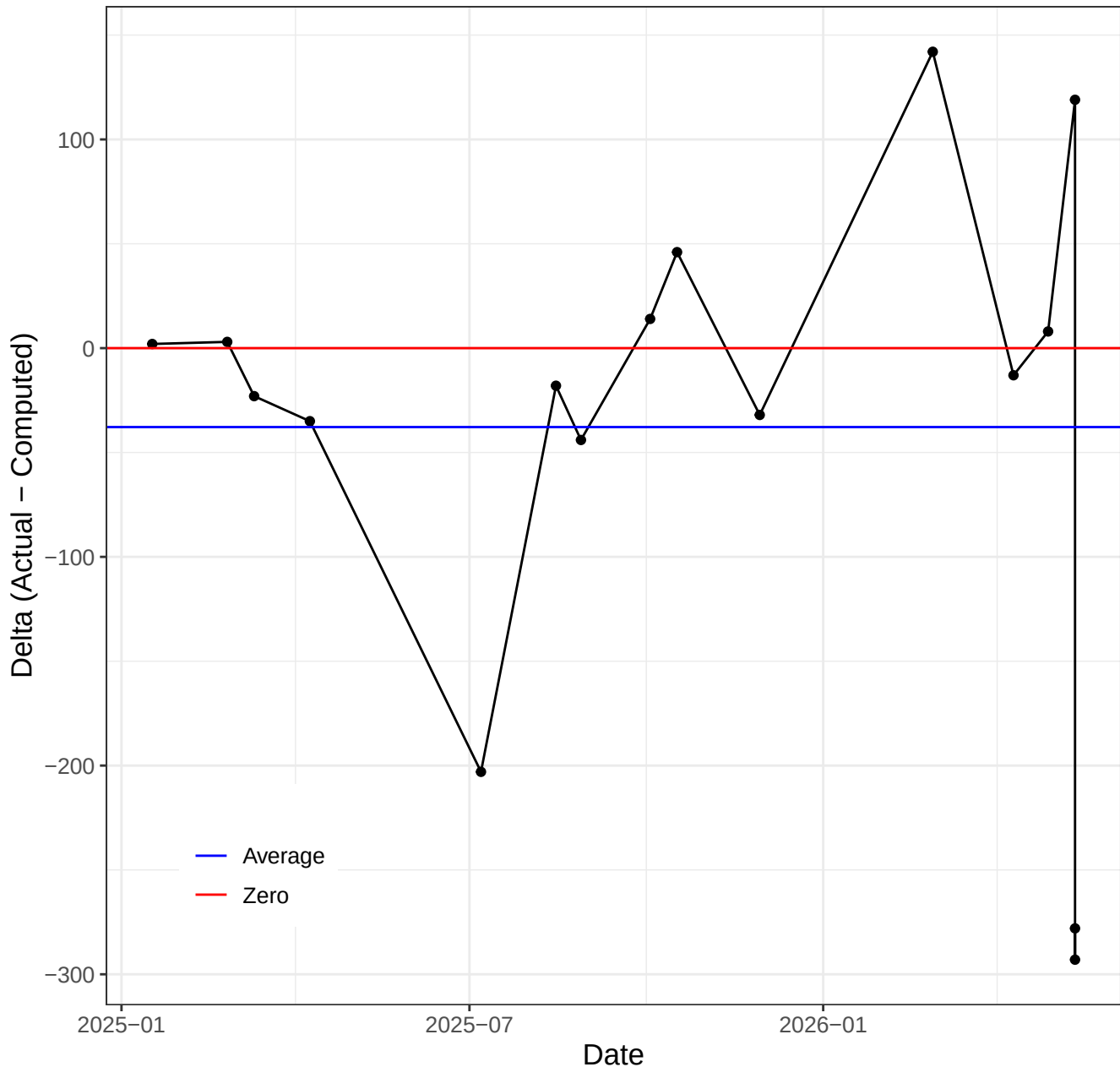
Here we can see again our average distance is much higher than our zero (0) on the y-axis. We should take note of how wide our 95% Limits are for this, it's about -250' to +1,000', a 1,250' range for our flights. This is massively different from our Ground Roll distances, and these points are better spread with fewer outliers, meaning there's a lot of variation in how the pilot performs. We also see the points start to get higher as they move closer to the right, which suggests the differences between computed and actual increases as the take-off distance increases. This wide range also reinforces our earlier *t-test* finding that the overrun for clearing a 50' obstacle is significant and somewhat unpredictable from flight to flight.

6 Deltas Over Time

Our last plots will be looking at the deltas as they happened over time. This will help us see if there's any trends in our deltas as time goes on, such as an engine wearing out, fluids needing to be changed, batteries needing to be recharged, or any other aircraft maintenance.

6.1 Ground Roll

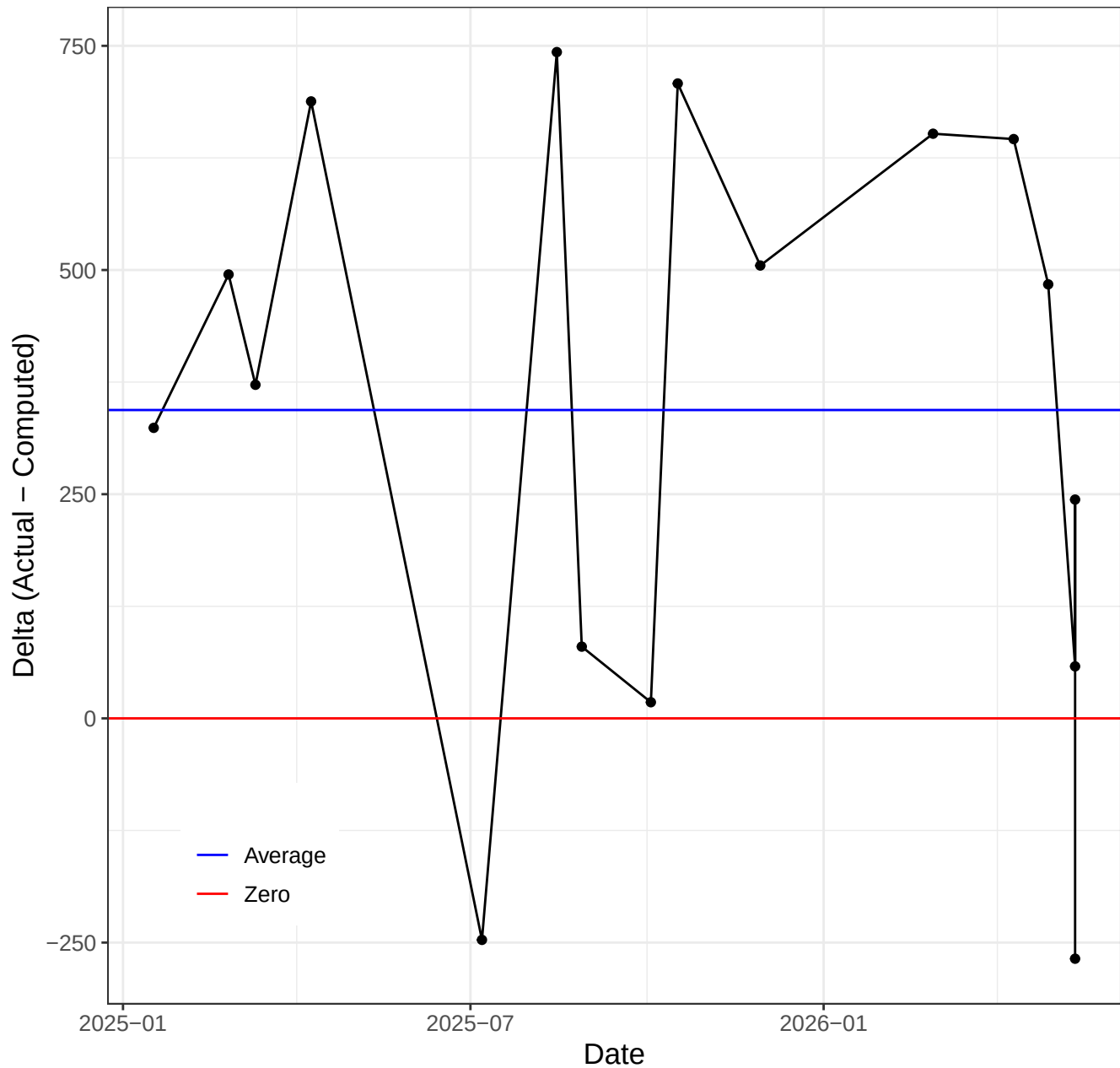
Ground Roll Delta Over Time



Here we see no clear trend between the points. We keep seeing the same points that have been standing out to us, with the rest staying pretty near the center. There's no strong evidence for any gradual degradation in performance.

6.2 50' Obstacle

50ft Clearance Delta Over Time



Here there is also no clear trend, but we can see how much variation is in each flight. Our deltas swing back and forth from -250' to +750' with no consistent pattern. We see a big drop around July (2025-07), which matches with our previous plot's Ground Roll dip, telling us this specific flight was unusual and outperformed predictions in both ground roll and 50' clearance. What's most important about this is we can see there's no trend saying our deltas changed over time. This can reassure us nothing is getting progressively worse, the distance required to clear a 50' obstacle just varies from flight to flight.

7 Conclusion

This study was performed to determine whether this particular pilot's take-off performance was aligned with the POH predictions.

In our Ground Roll numbers we saw no significant differences and can conclude the pilot does perform as predicted. For our 50' Obstacle numbers we had highly significant findings, determining the pilot consistently requires more distance than predicted, averaging about 343' more.

Since the ground roll is on target but the 50' clearance distance is not, the issue appears to be in the climb phase after lift-off. As illustrated in the introduction, the angle at which a pilot climbs after lift-off makes a significant difference in how quickly they can clear a 50' obstacle. This suggests the pilot's climb angle after lift-off may be worth reviewing, especially since the angle of lift-off is never computed or recorded.

There are no trends over time to let us know if these inconsistencies are getting worse or better, though further investigation into the climb performance may be worthwhile. Particularly recording what angles the aircraft takes off at and comparing them to the angles needed to reach the ideal, computed numbers for take-off.

Only having 16 flights and some incomplete records, this study could benefit from having more flight records to draw from to investigate the difference in clearing a 50' obstacle more in depth, particularly to investigate if it could be related to engine performance, climb technique, environmental conditions, etc.

Thank you for reading!

8 Tables Used

8.1 Data Dictionaries

Table 1: Flight Data Dictionary

Variable	Description
Date	Date of flight
Temp	Outside air temperature in Celsius
Press Alt	Pressure altitude, altitude in feet adjusted for atmospheric pressure
Den Alt	Density altitude, how high in feet the aircraft performs like it's at, accounting for temperature effects on air density
GW	Gross weight of the aircraft including pilot, passengers, and fuel
TO Dist	POH computed ground roll distance in feet before aircraft takes flight
50' Obs	POH computed total distance to clear a 50ft obstacle in feet
TO Roll	Actual recorded ground roll distance in feet before aircraft takes flight
50' Dist	Actual recorded total distance to clear a 50ft obstacle in feet
TO Roll Delta	Difference between actual and computed ground roll (Actual - Computed)
50' Delta	Difference between actual and computed 50ft clearance distance (Actual - Computed)

Table 2: POH Table Data Dictionary

Variable	Description
Weight (lbs)	Gross weight of the aircraft in pounds, table provides values for 2400, 2600, and 2800 lbs
lift-off Speed (KIAS)	Indicated airspeed in knots at which the aircraft lifts off the runway
Speed at 50ft (KIAS)	Indicated airspeed in knots at which the aircraft passes the 50ft obstacle
Pressure Alt (ft)	Pressure altitude in feet, ranges from sea level to 8000ft in the table
Ground Roll (ft)	POH predicted ground roll distance in feet for a given weight, altitude, and temperature
Total to Clear 50ft (ft)	POH predicted total distance in feet to clear a 50ft obstacle

8.2 Flight Records

Table 3: Flight Data Records

Date	Temp	Press Alt	Den Alt	GW	TO Dist	50' Obs	TO Roll	50' Dist	TO Roll Delta	50' Delta
2025-01-17	NA	2000	1314	2307	715	1275	717	1599	2	324
2025-02-25	11	2200	2045	2295	715	1225	718	1720	3	495
2025-03-11	20	2200	3300	2420	800	1410	777	1782	-23	372
2025-04-09	3	NA	NA	2489	730	1310	695	1998	-35	688
2025-07-07	25	2000	3520	2500	1000	1775	797	1528	-203	-247
2025-08-15	20	NA	NA	1925	770	1370	752	2113	-18	743
2025-08-28	13	NA	NA	2557	925	1650	881	1730	-44	80
2025-10-03	14	1700	2078	2411	745	1325	759	1343	14	18
2025-10-17	9	1900	1732	2411	715	1275	761	1983	46	708
2025-11-29	0	1615	-35	2360	665	1190	633	1695	-32	505
2026-02-27	8	2000	NA	2436	715	1275	857	1927	142	652
2026-04-10	14	1750	2046	2592	740	1325	727	1971	-13	646
2026-04-28	12	2000	2175	2592	860	1535	868	2019	8	484
2026-05-12	15	1800	2367	2313	738	1325	857	1383	119	58
2026-05-12	15	1800	2367	2313	738	1325	445	1569	-293	244
2026-05-12	15	1800	2367	2313	738	1325	460	1057	-278	-268

SECTION 5
PERFORMANCE

CESSNA
MODEL 177RG

TAKEOFF DISTANCE
MAXIMUM WEIGHT 2800 LBS

CONDITIONS:

- Flaps 10°
- 2700 RPM and Full Throttle Prior to Brake Release
- Mixture Set at Placard Fuel Flow
- Cowl Flaps Open
- Paved, Level, Dry Runway
- Zero Wind

MIXTURE SETTING	
PRESS ALT	GPH
S.L.	17
2000	16
4000	15
6000	14
8000	13

NOTES:

- Maximum performance technique as specified in Section 4.
- Landing gear extended until takeoff obstacle is cleared.
- Where distance value has been deleted, climb performance after lift-off is less than 150 fpm. Rate of climb is based on landing gear extended and flaps 10° at takeoff speed.
- Decrease distances 10% for each 9 knots headwind. For operation with tailwinds up to 10 knots, increase distances by 10% for each 2 knots.
- For operation on a dry, grass runway, increase distances by 15% of the "ground roll" figure.

WEIGHT LBS	TAKEOFF SPEED KIAS		PRESS ALT FT	0°C			10°C			20°C			30°C			40°C		
	LIFT OFF	AT 50 FT		GRND ROLL	TO CLEAR 50 FT OBS	TOTAL	GRND ROLL	TO CLEAR 50 FT OBS	TOTAL	GRND ROLL	TO CLEAR 50 FT OBS	TOTAL	GRND ROLL	TO CLEAR 50 FT OBS	TOTAL	GRND ROLL	TO CLEAR 50 FT OBS	TOTAL
2800	58	62	S.L.	795	1420	1530	860	925	1645	925	1010	1800	995	1765	1900	1065	1170	2090
			1000	870	1555	1675	935	1010	1800	1010	1105	1980	1085	1940	2090	1170	1280	2310
			2000	950	1705	1835	1025	1105	1980	1105	1210	2185	1190	2140	2310	1280	1405	2565
			3000	1040	1875	2025	1125	1210	2185	1210	1330	2425	1305	2365	2565	1405	1545	2860
			4000	1140	2065	2240	1230	1330	2425	1330	1460	2700	1435	2630	2860	1545	1700	3215
			5000	1250	2290	2485	1355	1460	2700	1460	1610	3030	1575	2945	3215	1700	1880	3645
			6000	1380	2550	2780	1490	1610	3030	1610	1780	3430	1740	3315	3645	1880	---	---
			7000	1520	2855	3125	1645	1780	3430	1780	1970	3925	1920	3780	---	---	---	---
			8000	1680	3225	3550	1815	1970	3925	1970	---	---	---	---	---	---	---	---

Figure 5-4. Takeoff Distance (Sheet 1 of 2)

TAKEOFF DISTANCE
2600 LBS AND 2400 LBS

REFER TO SHEET 1 FOR APPROPRIATE CONDITIONS AND NOTES.

WEIGHT LBS	TAKEOFF SPEED KIAS		PRESS ALT FT	0°C		10°C		20°C		30°C		40°C	
	LIFT OFF	AT 50 FT		GRND ROLL	TOTAL TO CLEAR 50 FT OBS	GRND ROLL	TOTAL TO CLEAR 50 FT OBS	GRND ROLL	TOTAL TO CLEAR 50 FT OBS	GRND ROLL	TOTAL TO CLEAR 50 FT OBS	GRND ROLL	TOTAL TO CLEAR 50 FT OBS
2600	56	60	S.L.	670	1200	720	1290	775	1380	835	1480	895	1590
			1000	730	1310	785	1405	845	1510	910	1620	975	1740
			2000	795	1430	860	1535	925	1650	995	1775	1070	1915
			3000	870	1565	940	1685	1010	1815	1090	1955	1170	2110
			4000	955	1720	1030	1855	1110	2000	1195	2160	1285	2335
			5000	1045	1895	1130	2045	1220	2215	1315	2395	1415	2600
			6000	1150	2095	1240	2270	1340	2460	1445	2675	1560	2910
			7000	1265	2330	1365	2530	1475	2750	1595	3000	1720	3280
			8000	1395	2600	1510	2835	1630	3100	1760	3400	1900	3740
			S.L.	560	1005	600	1075	645	1150	695	1230	745	1320
2400	54	57	1000	605	1090	655	1170	705	1255	755	1345	810	1440
			2000	665	1190	715	1275	770	1370	825	1470	885	1575
			3000	725	1300	780	1395	840	1495	905	1610	970	1730
			4000	790	1420	855	1530	920	1645	990	1770	1065	1905
			5000	865	1560	935	1680	1010	1810	1085	1950	1165	2105
			6000	950	1715	1025	1850	1105	1995	1190	2155	1285	2335
			7000	1045	1895	1130	2045	1215	2215	1310	2400	1415	2605
			8000	1150	2100	1240	2275	1340	2470	1445	2685	1560	2920

Figure 5-4. Takeoff Distance (Sheet 2 of 2)

9 Citations

Figure 1 (take-off Sequence Diagram): Federal Aviation Administration. (2021). *Airplane Flying Handbook* (FAA-H-8083-3C), Chapter 6: take-offs and Departure Climbs. U.S. Department of Transportation.

https://www.faa.gov/regulations_policies/handbooks_manuals/aviation/airplane_handbook

Figure 2 (V_x vs V_y Diagram): Pilot's Path. (2018). *Best angle vs best rate of climb* [Video]. YouTube.

<https://www.youtube.com/watch?v=EgtjSPXjgWQ>

P-Value Significance Levels: Cowles, M., & Davis, C. (1982). *On the origins of the .05 level of statistical significance*. *American Psychologist*, 37(5), 553–358.

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POH Performance Tables: Cessna Aircraft Company. (1976). *Pilot's Operating Handbook: Cessna Model 177RG Cardinal RG*, Section 5: Performance. Cessna Aircraft Company.

https://rafc.net.au/wp-content/uploads/2019/05/177rg_poh_1976_e.pdf